CURRICULUM VITAE

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RESEARCH INTERESTS

1. **Subseasonal-to-Seasonal Prediction and Predictability**
    - Predictability, forecast skill, and systematic biases in coupled prediction systems
    - Evaluation and improvement of predictions across weather-to-climate timescales
2. **Climate Dynamics, Teleconnections, and Extremes**
    - Tropical climate variability and tropical–extratropical interactions
    - Large-scale atmospheric circulation, teleconnection processes, and climate extremes
3. **Coupled Earth System Processes and Data Assimilation**
    - Atmosphere–ocean and land–atmosphere interactions
    - Coupled data assimilation and initialization of Earth system prediction models

EDUCATION

2015.03 – 2021.02	**Ph.D. in Environmental Science and Engineering**, UNIST, Korea **Dissertation:** *Development of a Coupled Data Assimilation System in the Fully Coupled Model and Its Implications for Seamless Prediction* **Advisor:** Prof. Myong-In Lee
2009.03 – 2015.02	**B.S. in Earth Science and Engineering**, UNIST, Korea

EMPLOYMENTS

2025.09 - Present	**Research Assistant Professor**, UNIST
2023.06 – 2025.09	**Postdoctoral Research Fellow**, George Mason University
2021.03 – 2023.06	**Postdoctoral Research Associate**, UNIST

RESEARCH EXPERIENCE

2023.01 – 2023.05	**Visiting Scientist**, Met Office Hadley Centre, UK
2017.02	**Visiting Researcher**, NASA GSFC, Climate and Radiation Lab, USA
2013.06 – 2013.08	**Internship**, NASA GSFC, GMAO, USA

PROFESSIONAL SERVICES

2024.10 – 2026.10	**Editor**, Korean Atmospheric Scientists in America (KASA)

PUBLICATIONS

[17] **Choi, N.**, Quan, X-W., and Stan, C. (2026). Drivers of the Asymmetric ITCZ Bias in Climate Models, Scientific Reports. (Accepted).

[16] Lee, J., Lee, M. I., Kim, J., Bak, G., **Choi, N.**, Lee, J., ... & Shin, B. (2026). Atlantic SST as a key source of subseasonal predictability in the April 2020 Siberian heatwave. npj Natural Hazards.

[15] **Choi, N.**, Stan, C., Lee, B., & Seo, E. (2026). Potential sources of predictability for the surface air temperature over the CONUS during boreal winter. Journal of Climate, 39(16), 5005-5016.

[14] **Choi, N.**, Stanczak, J., & Stan, C. (2026). The Role of Atmosphere–Ocean Coupling on the Prediction of Madden–Julian Oscillation. Journal of Climate, 39(13), 3669-3684.

[13] Kim, K., Lee, M. I., Lee, S., **Choi, N.**, Jin, F. F., Min, S. K., & Kam, J. Local and Remote Drivers of Increased Variability in Extreme Wildfire Conditions in Southeastern Australia. Available at SSRN 5708203.

[12] Lee, S., Lee, M. I., Lee, S., Jin, F. F., **Choi, N.**, & Tak, S. (2025). Distinct Features of the Summer Arctic Oscillation. Journal of Climate, 38(21), 5983-5996.

[11] **Choi, N.**, Lee, M. I., Ham, Y. G., Hyun, Y. K., Lee, J., & Boo, K. O. (2025). Reducing initialization shock by atmosphere-ocean coupled data assimilation and its impacts on the subseasonal prediction skill. Journal of Climate, 38(6), 1389-1401.

[10] **Choi, N.**, Lee, M. I., Tak, S., Cha, D. H., & Watanabe, M. (2025). Siberian heat extremes caused by Eurasian atmospheric teleconnections and amplified by local land surface conditions. Environmental Research Letters, 20(3), 034029.

[9] **Choi, N.**, & Stan, C. (2025). Large-Scale Surface Air Temperature Bias in Summer over the CONUS and Its Relationship to Tropical Central Pacific Convection in the UFS Prototype 8. Journal of Climate, 38(1), 117-129.

[8] Kim, H., **Choi, N.**, Lee, M. I., Tak, S., Cha, D. H., & Min, S. K. (2024). Causes on the 2018 record-breaking heatwave in South Korea: compound impacts from recurrent large-scale atmospheric teleconnections. Environmental Research Letters, 19(12), 124064.

[7] Tak, S., **Choi, N.**, Lee, J., & Lee, M. I. (2024). Probabilistic medium-range forecasts of extreme heat events over East Asia based on a global ensemble forecasting system. Weather and Climate Extremes, 45, 100694.

[6] **Choi, N.**, & Lee, S. (2023). Comparison of the opposite behaviours of Korean heatwaves with extreme hot sea surface temperatures in August 2016 and 2022. International Journal of Climatology, 43(15), 6993-7002.

[5] Lee, S., Park, M. S., Kwon, M., Park, Y. G., Kim, Y. H., & **Choi, N.** (2023). Rapidly changing East Asian marine heatwaves under a warming climate. Journal of Geophysical Research: Oceans, 128(6), e2023JC019761.

[4] Yasunari, T. J., Nakamura, H., Kim, K. M., **Choi, N.**, Lee, M. I., Tachibana, Y., & da Silva, A. M. (2021). Relationship between circum-Arctic atmospheric wave patterns and large-scale wildfires in boreal summer. Environmental Research Letters, 16(6), 064009.

[3] **Choi, N.**, Lee, M. I., Cha, D. H., Lim, Y. K., & Kim, K. M. (2020). Decadal changes in the interannual variability of heat waves in East Asia caused by atmospheric teleconnection changes. Journal of Climate, 33(4), 1505-1522.

[2] **Choi, N.**, Kim, K. M., Lim, Y. K., & Lee, M. I. (2019). Decadal changes in the leading patterns of sea level pressure in the Arctic and their impacts on the sea ice variability in boreal summer. The Cryosphere, 13(11), 3007-3021.

[1] **Choi, N.**, & Lee, M. I. (2019). Spatial variability and long-term trend in the occurrence frequency of heatwave and tropical night in Korea. Asia-Pacific Journal of Atmospheric Sciences, 55, 101-114.

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[5] Tak et al., Incremental Contribution of Soil Moisture Data Assimilation to Sub-seasonal Prediction of the 2022 Central Eastern China Heatwave. Weather and Climate Extremes. In Review.

[4] Sohn et al., K-R2O in Action: Enhancing Operational Reliability through the Independent Testbed for National Climate Prediction System. Bulletin of the American Meteorological Society. In Review.

[3] Xavier, P., Willett, M., ..., **Choi, N.**, ..., and Zhu, H.: Assessment of the Met Office Global Coupled model version 5 (GC5) configurations, Geoscientific Model Development Discussion. In Review.

[2] Lee et al., Understanding the physical mechanism of recording-breaking 2025 East Asian heatwave. Environmental Research Letters. In Review.

[1] Lee et al. Extratropical Rossby Waves Amplify Central Siberian Fire Risk Through Subseasonal Teleconnections, Agricultural and Forest Meteorology. In Review.

PATENT

- Medium-Range Heatwave Forecasting System and Method (Domestic, South Korea, 2019)

AWARDS

- Young Atmospheric Scientist Award, Korean Meteorological Society (2025)
- Outstanding Poster Presentation Award, Korean Meteorological Society (2019)